

Advanced Deep Learning Techniques for Anomaly Detection in Cloud Computing Traffic: Methods and Applications

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Abstract— To increase anomaly detection in complicated and imbalanced network settings, this study presents an Enhanced Convolutional Neural Network (ECNN) model. Using advanced techniques in deep learning, clustering, and optimisation, the objective is to build a robust and trustworthy system capable of accurately detecting anomalies. use the UNSW-NB15 dataset, which is well-known for its diverse network traffic statistics, to train and evaluate the model. Utilising K-Means++ for initial clustering, applying a deep auto-encoder for lowering dimensionality, and evaluating the processed data with an ECNN are some of the many processes that make up the methodology. Layers that make up the ECNN include convolutional, pooling, and fully connected ones. Training the model to handle class imbalance involves using a mix of cross-entropy and targeted loss functions. We use a Genetic Algorithm to optimize the hyperparameters. Metrics used to assess performance include F1-score, recall, accuracy, and precision. With an F1 score of 99.75%, recall of 98.78%, precision of 99.56%, and accuracy of 99.85%, the ECNN model performs admirably. The model's exceptional accuracy and recall in detecting anomalies validates its performance in managing complicated network traffic circumstances, according to the results.

Keywords—Enhanced Convolutional Neural Network (ECNN), Anomaly detection, Intrusion detection, UNSW-NB15 dataset and Hyper parameter optimization

I. INTRODUCTION

Modern information technology infrastructures would not be complete without cloud computing, which has allowed businesses to grow, become more flexible, and better manage their resources in this era of digital transformation. With the increasing complexity and scale of cloud environments, it has become crucial to prioritize the security and performance of these systems. An urgent issue in this field is identifying abnormal network traffic, which can indicate security risks like cyber-attacks, data breaches, or system failures. Conventional techniques for identifying unusual events frequently have difficulties in keeping up with the constantly changing and large amount of data flowing through cloud-based networks[1]–[4]. This shortcoming highlights the need for more sophisticated and flexible technology to enable rapid and precise anomaly identification. When it comes to solving difficult cybersecurity problems, deep learning has been a game-changer. among others. When compared to more conventional approaches to anomaly identification, such as

those based on statistical criteria or predefined rules, deep learning algorithms perform far better. They excel at delving into massive datasets, finding hidden patterns, and drawing insightful conclusions. The remarkable effectiveness of neural networks and other algorithms in analysing high-dimensional data makes them ideal for identifying intricate patterns in network traffic. There are a lot of advantages to using deep learning to spot suspicious activity in cloud computing networks. When it comes to handling data pertaining to network traffic, which is complicated and multidimensional, deep learning models work wonders. The complexity and unpredictability of cloud communications make it difficult for conventional methods to handle them, which increases the likelihood of false positives and missed detections. Deep learning models excel at detecting anomalies in traffic patterns by utilizing their capacity to learn hierarchical representations of data, resulting in enhanced accuracy compared to other methods[5]–[7].

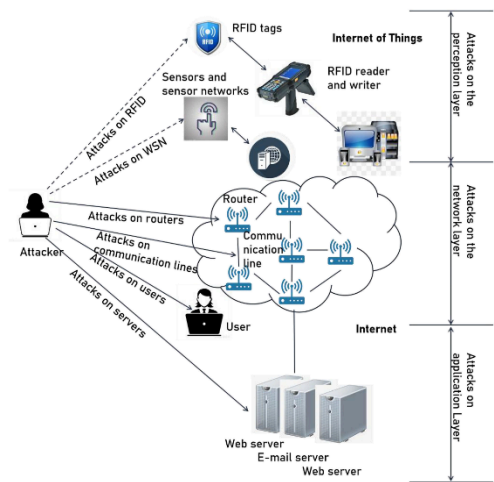


Fig. 1 Anomaly Traffic Detection in Cloud Computing Environment

Anomaly detection solutions based on deep learning provide improved flexibility and scalability. Cloud environments are distinguished by their dynamic nature, where traffic patterns may fluctuate as a result of factors such as user behavior, application upgrades, and scaling processes. Continuously learning and adapting from new data is what makes deep learning models so efficient at responding to changing traffic patterns and consistently maintaining a high level of detection

performance over time. The capacity to adapt is essential for ensuring efficient security measures in a swiftly evolving cloud environment. Another significant benefit of deep learning is its capacity to enhance the effectiveness of anomaly detection. Conventional approaches can necessitate significant manual calibration and rule-based modifications, which can consume a substantial amount of time and are susceptible to mistakes. On the other hand, deep learning models have the ability to automatically extract features and recognize patterns, which greatly reduces the requirement for user involvement and allows for the immediate detection of anomalies[8]–[11]. Not only does this make security professionals' jobs easier, but it also speeds up the process of threat detection. But there are also special challenges associated with using deep learning for anomaly traffic identification. Certain organisations may find it challenging to train deep learning models due to the massive computational resources and large amounts of high-quality data that are required. Ensuring the security and privacy of sensitive information used for training is of the highest priority. To protect the data from misuse or unauthorised access, this is of the utmost importance. Just because deep learning models are better at detection doesn't mean they're foolproof. Their efficacy is dependent on the data's quality and the training procedure's robustness. Preserving the models' accuracy and reliability in real-world circumstances requires careful attention to training and verification. Using deep learning to identify unusual traffic in cloud computing settings is a huge step forward in protecting networks. Organisations can achieve better accuracy, flexibility, and efficiency in detecting suspicious network activity by utilising deep learning algorithms. As a result, their cloud-based systems become more secure and functional. There will likely be an increasing reliance on deep learning to address the security issues raised by cloud computing as it develops further. As a result, the field will see an uptick in innovation and new discoveries [12]–[15].

II. RELATED WORK

Zuo 2024 et al. An essential part of maintaining cyberspace secure is the ability to identify anomalies in network traffic by studying network behaviour for signs of malicious attacks. Traditional machine learning methods already struggle to manage enormous datasets, and the exponential rise in network traffic due to fast developments is making the problem even worse. A better option would be to use deep learning, which can generalise and extract traits. An improved variant of the Osprey optimisation algorithm and a Deep Residual Shrinkage Network (DRSN) form the basis of the "GSOOA-1DDRSN" method, which is detailed in this study. The declared aims of the approach are dimensionality reduction and essential feature selection. Results from testing on the NSL-KDD and UNSW-NB15 datasets show that GSOOA-1DDRSN improves classification accuracy, precision, recall, and F1 Score by 2% while decreasing computation time by 20% to 30% [16].

Wang 2024 et al. New developments in automotive networking have exacerbated security issues with the vulnerable Controller Area Network (CAN) bus. Using the bzip2 compression technology, this study introduces SIDuBzip2, a method for anomaly detection in communications. In order to detect abnormalities, the system takes advantage of the pseudo-periodic nature of CAN transmission by translating CAN ID values and computing similarity. The testing results show that compared to earlier methods, SIDuBzip2 is much faster and more accurate at identifying a broad variety of attacks [17].

Vibhute 2024 et al. New security threats have emerged alongside the developments in computer networking that have made data sharing between systems easier. Presenting a high-level Convolutional Neural Network (CNN) model, this research aims to detect and classify network intrusions in a non-flat cloud environment in near-real-time. When selecting features, the CNN employs a random forest model. Using the CSE-CIC-IDS2018 datasets, the CNN model achieved an accuracy rate of 97.07% with a margin of error of only 2.93%. Furthermore, it served F1-scoring, recall-and precision-focused real-time Industry 4.0 systems well [18].

Javaheri 2023 et al. Distributed denial of service (DDoS) attacks are a common cybersecurity risk for organisations that depend on IT. Regardless of the presence of several intrusion detection systems, distributed denial of service attacks continue to cause significant financial and reputational impact. By comparing and contrasting prior research and offering a hierarchical structure, this paper aims to give a thorough evaluation of distributed denial of service assaults. To fix the system's present flaws, the paper proposes effective defensive measures, with an emphasis on fuzzy-based detection methods. The results can help businesses and government agencies better prepare for cybersecurity threats and ensure continued operations [19].

Yue 2022 et al. an approach to network security that exploits open-source big data technologies to detect anomalous behaviour. This configuration combines two algorithms: EWMA and anomalous flow interval mapping. The testing approach includes checking the physical cluster settings, system performance, and DDoS attack verification using NSFOCUS anti-DDoS logs and traffic from the Tianjin Education Metropolitan Area. Unusual traffic patterns, flow direction, and event prioritisation are all areas where the platform excels. Its rapid error recovery and high fault tolerance make it a great option for real-time anomaly detection and dynamic network monitoring [20].

TABLE 1. LITERATURE SUMMARY

Author/Year	Method/Model	Research Gap	Parameters
Nithesh/2022 [21]	CNN Model	Existing methods lack automated anomaly detection.	Accuracy of 82%

Yao/2022 [22]	DCNN Model	Current models lack efficiency in detecting DoS attacks in WSNs.	Accuracy of 97.83% Recall 97.83 Precision 98.02 F1-score 97.92
Ming/2022[23]	CNN Model	Deep learning models struggle with imbalanced datasets; need efficient solutions.	Accuracy of 99.99%
Hwang/2020 [24]	CNN Model	Existing defenses lag in real-time anomaly detection	Accuracy of 82%

III. METHODOLOGY

Our proposal presents a novel method for identifying anomalies by utilising deep learning in conjunction with robust clustering and optimisation approaches. With this, intrusion detection systems will be able to handle complicated and unbalanced environments better. We employ the UNSWNB15 dataset, which is the latest and full library of network traffic data. It is considered perfect to create and to test models robustly because it covers a wide range of events related to a network such as ordinary activities and malicious activities. In order to guarantee consistencies for scalability and enhance accuracy, performance, and convergence times during training, deep learning models preprocess the standardization of features to maintain equality among each of them. The first step is K-Means++ clustering, which is superior to its predecessors in the initialization of centroids and consequently enhances clustering accuracy. Afterwards, a deep auto encoder lowers the dimensionality of the dataset while preserving important information, which improves data management and the efficiency of model training. The ECNN combines multi-convolution layers, pooling, and fully connected layers to extract complex patterns and spatial hierarchies from input data. All of these methods improve the training of the ECNN in terms of cross-entropy and targeted loss functions, given that the latter fits the minority classes and easing the class imbalance problem effectively, which is quite critical for anomaly detection scenarios given that the instances of normal data are more than those of the abnormal ones. It includes hyper-parameters like learning rate, batch size, and network architecture parameters. In this respect, the current work employs a Genetic Algorithm that can optimize such highly relevant parameters concerning detection performance by an adaptive tuning process. We validate the proposed technique on the UNSW-NB15 dataset and make use of comprehensive evaluation metrics like F1-score, recall, accuracy, and precision in order to assess the model's efficiency and its capability to detect outliers and abnormal network activities.

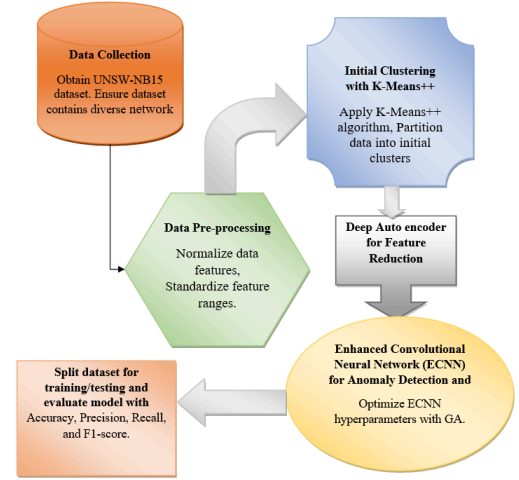


Fig. 2 Proposed Flowchart

A. Data Collection

In order to train and evaluate intrusion detection algorithms, this research makes use of the UNSW-NB15 dataset, which offers a complete set of up-to-date network traffic data. There are examples of both healthy and malignant network activity in this dataset. This tool is absolutely crucial while working on anomaly detection models. By utilising the diverse and extensive data available in UNSW-NB15, we can guarantee strong model performance and precise evaluation of intrusion detection capabilities. Thanks to the comprehensive and up-to-date dataset attributes, intrusion detection research has progressed towards more effective processing of data with multiple dimensions and imbalanced classes.

B. Data Pre-processing

Data preprocessing ensures consistency in data size and quality, which is essential for effectively training deep learning models. Techniques like data normalization adjust data ranges to ensure uniformity across features, preventing any single feature from disproportionately influencing the model's training process. This stage is vital for managing high-dimensional datasets, enhancing model performance, and accelerating the convergence of learning algorithms. Additionally, data standardization improves training accuracy and efficiency, ultimately enhancing the model's detection capabilities.

C. Initial Clustering with K-Means++

Partitioning the dataset into k initial clusters is the first stage in first-pass clustering using K-Means++. This method improves the K-Means clustering algorithm's initial cluster centroids, which in turn improves the overall performance of clustering. K-Means++ improves clustering results and accelerates convergence by using a more strategic approach to finding initial centroids. The methodology's end result is clusters that are more precise and consistent, which lays a good groundwork for future studies and training of models. The enhanced initialisation considerably enhances the clustering process's efficiency and effectiveness, which is crucial when dealing with high-dimensional data.

D. Deep Autoencoder for Feature Reduction:

Training the model to decrease the dimensionality of the clustered data is the feature reduction technique that employs a deep autoencoder. A type of model known as an autoencoder may take in input that is high dimensional and reduce its dimensions while keeping all the important information. We need to simplify the data structure while keeping essential patterns and features in order to successfully manage the high-dimensional components of the information. Data processing and interpretation are both made easier by the autoencoder's reduction of the number of dimensions, which enhances computational efficiency and model performance. For deep learning model training to be efficient and effective, this data preparation stage is crucial.

E. Enhanced Convolutional Neural Network (ECNN) for Anomaly Detection:

Importing the reduced-dimensional data into the network is the first stage in Anomaly Detection using an ECNN. An ECNN architecture can recognise complicated input patterns thanks to its multiple convolutional, pooling, and fully connected layers. Integrating cross-entropy loss with focal loss during ECNN model training is the most effective way to handle class imbalance. In an effort to improve the model's performance on classes that are underrepresented, the targeted loss function gives hard-to-classify data priority. The ECNN's enhanced detection capabilities are a result of its meticulous design and concentrated training approach, which ensure dependable anomaly detection.

F. Parameter Optimization with Genetic Algorithm (GA)

Optimizing the hyperparameters (learning rate, batch size, number of layers, etc.) of an ECNN using a Genetic Algorithm (GA) is known as parameter optimization. The goal of the Genetic Algorithm (GA) is to find the best possible setup by iteratively refining these parameters. By simulating natural selection over many generations, the genetic algorithm (GA) determines which parameter combinations provide the best results. This process ensures that the ECNN is fine-tuned to reach maximum efficiency and accuracy, which improves its ability to detect outliers. To improve model performance in complex, high-dimensional scenarios, the genetic algorithm is a powerful tool due to its excellent optimisation abilities.

G. Experimental Validation

To evaluate the model's performance, the UNSW-NB15 dataset is partitioned into training and testing subsets. This methodology ensures a clear separation between the training and evaluation phases, thereby mitigating the risk of overfitting. Key performance metrics, including F1-score, recall, accuracy, and precision, are calculated to provide a comprehensive assessment of the model. These metrics facilitate the evaluation of the model's effectiveness, the identification of dataset anomalies, and its capability to recognize diverse patterns in network traffic.

Hyperparameter	Values
Learning Rate	0.001, 0.01, 0.1
Batch Size	32, 64, 128
Number of Epochs	100,
Number of Convolutional Layers	3, 4, 5
Number of Filters	32, 64, 128
Activation Function	ReLU, Sigmoid
Dropout Rate	0.2, 0.4, 0.5

IV. RESULT & DISCUSSION

A. Accuracy

As a performance metric, accuracy in a dataset is defined as the percentage of correctly predicted instances relative to the total number of occurrences. It is calculated by summing all the true positive and true negative predictions and dividing this total by the number of samples. While accuracy provides a general measure of performance, it may not fully represent a model's reliability, particularly when dealing with imbalanced datasets where the proportion of positive and negative instances is significantly unequal.

Relying solely on accuracy in such cases can be misleading, as a model could achieve high accuracy by predicting only the majority class. To address this limitation and provide a more comprehensive evaluation, additional metrics such as recall, precision, and F1-score are necessary. These metrics ensure a balanced assessment of the model's ability to handle imbalanced datasets and deliver accurate predictions across all classes.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

B. Precision

One way to quantify the accuracy of a model is by looking at its precision, which is defined as the ratio of correctly predicted positive outcomes to all projected positive events. After tallying up all the positive and negative outcomes, divide the sum by the total number of right answers to obtain the final result. Accuracy is critical for high false positive costs since it guarantees that the instances recorded as positive are true positives. If an intrusion detection system has a high level of accuracy, it should have little trouble differentiating between legitimate and malicious activities. Precision is a crucial statistic to examine in situations when false alarms could lead to significant consequences in order to evaluate the dependability of the model's positive predictions.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

C. Recall

Recall estimates the number of actual positives that the model accurately detected; it is also known as sensitivity or true positive rate. After tallying up all the positive and negative outcomes, divide the sum by the total number of right answers to obtain the final result. Recollection is crucial, especially

TABLE 2 HYPERPARAMETER TABLE

because the consequences of omitting a positive example could be catastrophic. Missing an intrusion while searching for security vulnerabilities, for instance, could have catastrophic consequences. With a high recall score, the model is able to accurately identify and include the majority of positive examples. Improving memory could lead to worse precision, therefore it's important to strike a balance between recall and other measures.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

D. F1 Score

When evaluating a classification model, the F1-Score is a useful metric to keep in mind. It provides a fair assessment of the model's efficacy since it considers both recall and precision. To establish a happy medium between recall and precision, the F1-score integrates the two into a single statistic in mathematics. A two-fold multiplication of the "Precision" and "Recall" fractions by " $\frac{2}{\frac{1}{precision} + \frac{1}{recall}}$ " completes the calculation. Because it accounts for both false positives and false negatives, the F1-score is useful when there is a mismatch in the distribution of classes. An enhanced F1-score, a measure of comprehensive performance, indicates that the model achieves a solid balance between recall and precision. It really comes into its own when the impact of both false positives and false negatives is substantial.

$$F1\ score = \frac{2}{\frac{1}{precision} + \frac{1}{recall}} \quad (4)$$

TABLE 3. PERFORMANCE EVALUATION OF PROPOSED MODEL ECNN

Models	Accuracy	Precision	Recall	F1 score
Proposed ECNN	99.85	99.56	98.78	99.77

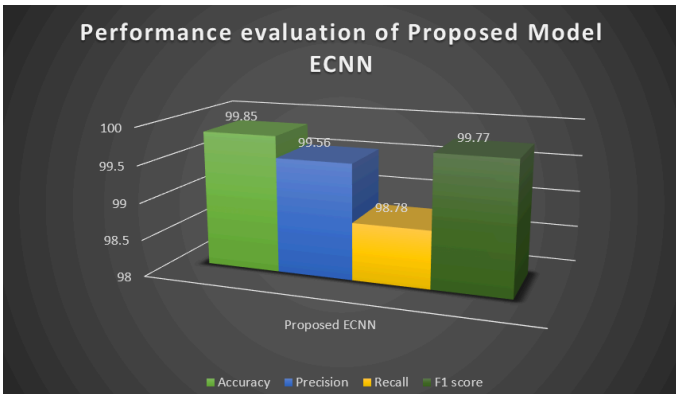


Fig. 3 Performance Graph of proposed Model

Table 3 presents the evaluation of the proposed ECNN model's efficiency, demonstrating outstanding outcomes in critical parameters. The model's remarkable accuracy rating of 99.85% allowed it to accurately classify almost all occurrences. With an accuracy score of 99.56%, the model

clearly excels at predicting real positives while producing very few false negatives. The model effectively decreases the number of missed detections while correctly detecting a substantial majority of actual positive cases, boasting a recall rate of 98.78%. A result of 99.77% indicates an accurate and precise performance, as the F1 score combines great recall and precision into one statistic. Taking everything into account, the results demonstrate that the ECNN model excels at identifying anomalies.

TABLE 4. COMPARATIVE ANALYSIS OF PROPOSED MODEL WITH EXISTING MODELS

Models	Accuracy	Precision	Recall	F1 score	Ref.
LeeNET (MLNET)	98.1	98.9	98.1	98.1	[14]
(SSAE) with (LSTM)	86.31	86.77	85.65	86.30	[25]
Proposed ECNN	99.85	99.56	98.78	99.77	

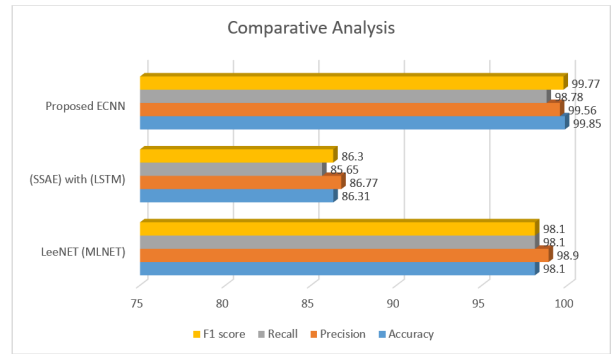


Fig. 4 Comparative Analysis Graph

To compare the ECNN model to other existing designs, Table 4 includes all of the relevant information. Measures of recall, accuracy, precision, and F1 score form the basis of the comparison. With an F1 score of 98.1%, recall of 98.9%, accuracy of 98.1%, and precision of 98.1%, LeeNET (MLNET) is an impressive network. A combination of the Sparse Stacked Autoencoder (SSAE) and the Long Short-Term Memory (LSTM) model produced the following performance metrics: recall—85.65%, accuracy—86.31 percent, precision—86.77 percent, and F1 score—86.30 percent. These measures show that performance is not as effective. On the other hand, the proposed ECNN model outperforms all previous models. A 99.75% F1 score, 98.78% recall, 99.56% precision, and 99.85% accuracy are all within its capabilities. By consistently outperforming rival models across the board, the ECNN model has proven its ability to properly detect outliers and categorise data from the given environment. In comparison to the other models tested, ECNN gets a high F1 score and general resilience by effectively reducing the prevalence of false positives and false negatives, respectively.

V. CONCLUSION

With respect to recall, accuracy, precision, and F1 score, the suggested Enhanced Convolutional Neural Network (ECNN) model outperforms state-of-the-art models like LeeNET and

SSAE with LSTM when it comes to anomaly detection. The methodology utilises the large-scale UNSW-NB15 dataset and applies state-of-the-art methods including K-Means++ clustering for improved centroid initialisation, deep autoencoders for dimensionality reduction, and evolutionary algorithms for hyperparameter optimisation. These upgrades improve the system's stability and dependability by addressing data imbalance and complex pattern concerns. The ECNN model achieved remarkable results with a recall of 98.78%, an F1 score of 99.77%, accuracy of 99.85%, and precision of 99.56%. With such a low rate of false positives and negatives, these results demonstrate how effectively it can classify data and identify outliers. The model is a powerful resource for improving cybersecurity protocols, and the findings suggest it has the potential to significantly improve intrusion detection systems that operate in real-time. This comprehensive study confirms the ECNN's function as an essential component of network traffic analysis for anomaly discovery by highlighting its strength and reliability.

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